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An empirical examination of the influence of biased personalized product recommendations on consumers' decision making outcomes☆,☆☆

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ABSTRACT

To assist consumers in product search and selection while shopping online, many e-commerce retailers have implemented web-based product recommendation agents (PRAs). However, consumers are empowered to the extent that the PRAs provide *true personalization* by recommending products based solely on, and thus best representing, consumers' preferences. This study constructs and empirically tests a theoretical model that examines how *biased recommendations* from PRAs influence consumers' *decision quality* and *decision effort*. The results of an online experiment show that consumers are extremely vulnerable to biased personalized recommendations from online PRAs. In addition, our results extend prior research by identifying perceived personalization as a critical mechanism driving the influence of biased PRA on consumers' decision quality and decision effort. This study fills a void in the literature and calls attention to an insidious form of manipulation made possible by innovative technologies supporting e-commerce.

1. Introduction

Today's digital marketplaces offer consumers great convenience, immense product choice, and large amounts of product-related information. However, as a result of the cognitive constraints of human information processing, finding products that satisfy consumers' needs and/or interests is not an easy task. Many online stores have implemented technological tools to assist consumers in product search and selection. One such tool is web-based product recommendation agents (PRAs), a type of web personalization technology that provides individual consumers with product recommendations based on their product-related needs and preferences expressed [1–3]. PRAs have the potential to help consumers make effective purchase decisions [4,5] and reduce consumers' uncertainty about buying products online [6].

Personalized recommendations are an e-commerce feature that is highly valued by consumers and also help online companies create brand loyalty [7]. This explains the phenomenal increase in the adoption of PRAs by online merchants in various industries (e.g., Amazon, eBay, Apple) [8]. Despite the potential for PRAs to aid consumers in decision making, consumers are empowered to the extent that the PRAs provide *true personalization* by recommending products based solely on and thus best represent consumers' preferences [9,10]. Just as a salesperson may simultaneously serve two principals (i.e., the client and the employing company) [11,12], a PRA may also be designed to serve the interest of its users (i.e., consumers) as well as that of its provider, creating opportunities for competing loyalties [11]. The objectives and economic incentives of the PRA providers (e.g., online merchants) are not always necessarily aligned with those of the consumers [13]. To balance their competing goals of aiding consumers in decision making and generating more business opportunities [8,14], online merchants may implement PRAs to provide recommendations that are not solely preference-matched to benefit consumers but instead are biased toward their own interests [3,8,15]. This may result in recommendations for products from certain vendors and/or with certain characteristics (e.g., high profit-margin products, soon-to-be discontinued products) so as to attain higher-than-usual profits (in terms of mark-ups, commissions, referral fees, etc.) [16,17] or reduce losses.

There have already been reported incidents of online companies (e.g., Amazon, Hotel.de) using PRAs to provide biased recommendations to consumers [14,18,19]. Prior academic research has also observed that PRAs have the potential to not only assist consumers but also steer them in a particular direction (e.g., [20–24]). However, to our knowledge, there has been limited understanding of the underlying mechanisms by which biased recommendations influence consumers'

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☆☆ This is a part of a duo of papers based on two *different* experimental studies, with this paper investigating the effect of biased recommendations by PRAs on consumer decision making, and the other paper exploring the design of interventions that can support consumers in detecting biased recommendations by PRAs.

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decision making outcomes in online buying, which is a gap the current study aims to fill.

Drawing on the theoretical perspective of personalization, this study examines how *biased personalized* product recommendations from PRAs influence consumers' *decision quality* and *decision effort* when shopping online. It further explores the role *perceived personalization* plays in mediating the impact of biased PRAs on consumer decision making. Specifically, this study compares the relative impact of *biased personalized* recommendations with that of *biased non-personalized* recommendations (based on false claim of popularity) on consumers' *decision quality* and *decision effort*. We posit that the availability of a PRA on a website may enhance a consumer's belief that due to personalization (be it true or not) her product search would be more effective and efficient. If the *biased personalized* recommendations do not result in improved objective decision quality for the consumer, while at the same time these recommendations lead the consumer to *believe* that her product choice is an optimal one, leading to her continued reliance on the PRA, then unscrupulous online merchants will have a powerful means with which they can influence consumer decision making toward the merchants' own benefits. With its focus on undesirable consequences of PRA use, this study answers the call for more academic research into the dark side of information technology (IT) [25].

The remainder of this paper is organized as follows. Section 2 presents the study's theoretical foundation and develops the hypotheses. The research method and results of hypothesis testing are reported in Sections 3 and 4. And the paper concludes with a discussion of the results, limitations, and contributions of the study and suggests areas for future research.

2. Theoretical foundations and hypotheses development

This section first reviews the theoretic perspective of personalization, which lays the theoretical foundation for this study. It then presents our research model and introduces associated hypotheses.

2.1. Theoretical perspective of personalization

While the idea of tailoring products and services to consumers' needs existed long before the inception of the Internet ([26]), advances in Internet technologies have made it possible to effectively offer personalized products and services on a grander scale. Personalization is defined as the "use of information about a particular user that provides tailored or personalized services for the user" ([27], p. 1). Using a consumer's demographics, shopping history, and preference information [1], websites with PRAs can generate recommendations that are personalized to her needs.

Prior research reveals that personalized recommendations have significant power in influencing consumers' decision making [28,29]. They are more accepted by consumers than non-personalized recommendations (i.e., those generated without consideration of consumers' personal preferences) [63], lead to more clicks than random suggestions [30], and result in higher user ratings when compared to random recommendations [31]. In mobile commerce, they lead to a significant increase in items viewed/sold and overall sales [32].

The power of personalized recommendations in influencing consumer decision making may be explained by the high level of *perceived personalization*, defined as the extent to which a consumer believes that a personalization system understands and represents her personal preferences [33]. Komiak and Benbasat [33] found that a PRA perceived by consumers to better understand their product related preferences and needs (i.e., is more personalized) enhanced their perception of the consensus between their own decision making and that of the PRA, and encouraged them to accept the recommendation of the PRA. Liang, Li, and Turban [34] noted that personalized recommendations induced intimate feelings (consisting of caring, commitment, communication, comfort, and trust) in consumers, which in turn enhanced consumers' attitude toward the recommendations.

Perceived personalization by a PRA can also positively shape consumers' attitudes toward the PRA itself and the websites implementing it. For instance, Zhang and Curley [35], as well as Komiak and Benbasat [33], found that perceived personalization significantly increased consumers' intention to adopt PRAs by increasing their cognitive trust and emotional trust in such systems. Kumar and Benbasat [36] showed that the provision of personalized recommendations increased both the perceived usefulness and social presence of the e-commerce website. A longitudinal experiment by de Pechpeyrou [30] also showed that consumers' attitude toward personalization positively influenced their attitude toward the website implementing personalized merchandising. Li [64] further noted that it is *perceived* personalization, rather than *actual* personalization, that explained message effectiveness.

Personalization is essentially system-driven [37], as consumers typically do not have high control over the actual inputs requested by the PRA (e.g., consumer preferences specifically considered by the PRA), its processes (e.g., algorithm for generating recommendations), or output (e.g., list of recommendations presented). This creates ambiguity regarding the veracity of the personalization: i.e., whether the recommendations provided by the personalization system are indeed generated solely on the basis of the consumers' preferences.

Extant research on personalization systems have been largely focused on their consumer benefits [1,3,15]. The current study addresses this gap in literature by examining the impact on consumer decision making of a personalization system (PRA, to be more specific) designed to bias consumers into choosing products that they would not otherwise have considered without using such system. By unravelling a potential mechanism, namely perceived personalization, that underlies the impact of biased PRAs, it adds to the limited body of research on the dynamics and effects of PRA bias, a type of strategic manipulation performed on the *generation* of information [15].

2.2. Research model

This study examines how consumers' decision making outcomes are influenced by biased product recommendations when shopping online. We are particularly interested in examining how a user's perception of personalization biases consumer judgment and decision making above and beyond order effect. Fig. 1 illustrates our Research Model, based on which a set of testable hypotheses are developed in Section 2.3.

2.2.1. Type of recommendations

This study compares consumers' decision making outcomes when they receive (1) biased *non-personalized* recommendations, or (2) biased *personalized* product recommendations from a PRA. We are particularly interested in examining how perceived personalization underlies the effect of biased PRAs on consumer judgment and decision making.

In the "***biased non-personalized*** recommendations" condition, a *sub-set of* available products that are promoted or "pushed" by the online store[1] are displayed before other, non-promoted products on the e-commerce website, with the false claim that the products are sorted in best-selling (or popular) order.

In contrast, in the "***biased personalized*** recommendations" condition, a *sub-set of* available products promoted or "pushed" by the online store are recommended by a PRA to consumers and displayed on the e-commerce website before other non-promoted products.[2] Within both sub-sets (i.e.,

[1] In the current study, *promoted products* refer to products for which biased recommendations are provided to enhance their attractiveness to consumers. They may not be the true optimal choice for consumers.

[2] While biased recommendations can be generated in various ways, this study focuses on PRAs that recommend products the website intends to promote over those that best fit consumers' preferences, so as to enhance the likelihood that the promoted products are chosen. This is akin to manipulations in personal selling situations, in which salespersons are motivated to recommend products that can bring them higher commission or products they are pressured to sell (e.g., products having a high profit margin) [16,17], rather than recommending products that best match customers' preferences.

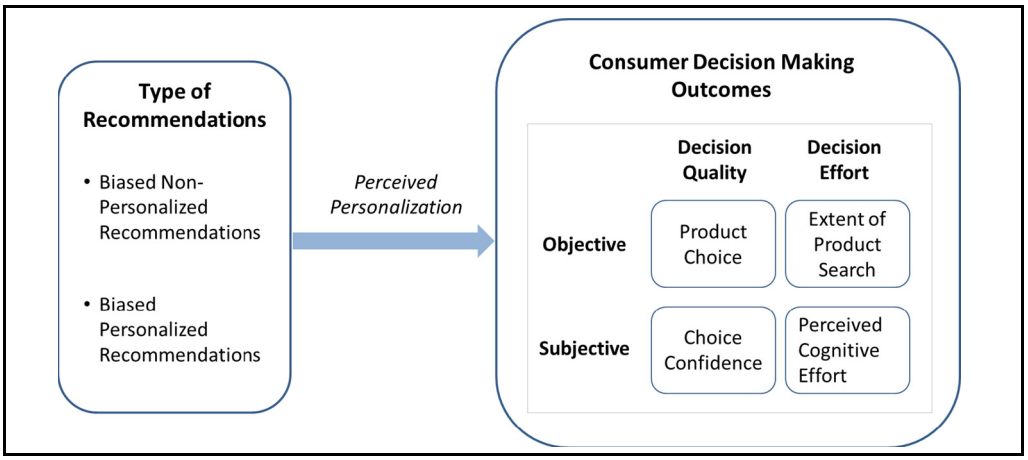


Fig. 1. Research model.

Table 1
Experimental manipulation.

Biased non-personalized recommendations	The 48 *promoted* products are randomly placed in the first 8 products pages whereas the 48 *dominant* products are randomly placed in product pages 9–16. The same order is shown to all users. There are a total of 16 product pages, with 6 products on each page The website states that all the products are ordered by popularity.	
Biased personalized recommendations	The website features a PRA that uses a set of needs-based questions (adapted from [3,61]) to obtain users' product-related preferences (see Appendix A). The PRA then calculates a "fit score" for every available product based on the importance weight (elicited from an individual user) and the gap between the user's requirement and the attribute level the alternative can satisfy (in accordance with the fit score calculation method described by Wang and Benbasat [62]). There are a total of 16 product pages, with 6 products on each page. The PRA employs an Additive Compensatory decision strategy rather than an Elimination by Aspects strategy (see [62]). Thus, it is possible that the most highly ranked cameras are not from the brands preferred by Pat (see Appendix A).	The 48 *promoted* products are placed in the first 8 product pages and ordered by how each product matches users' expressed preferences. The 48 *dominant* products are placed in the next 8 product pages (i.e., pages 9–16) ordered also by how each product matches users' expressed preferences. The website states that all the products are ordered by how they match consumers' expressed preferences (indicated by the products' fit scores). In expressing the product-related preferences, users need to mentally "translate" the set of Pat's camera-related preferences provided to them (see Appendix A) into concrete level of camera features (e.g., 10 mega-pixel; 5× optical zoom), as well as to indicate the importance of each feature. The variability in users' translation and their weighting of features will result in variability in the fit scores of their chosen products.

promoted and non-promoted), products are arranged in descending order of how well they fit customers' needs obtained in the PRA's preference-elicitation process (see Section 3.1 and Table 1 for full details).

2.2.2. Consumer decision making outcomes

In line with the taxonomy of decision effectiveness proposed by Lilien et al. [38] and following the practice of prior research assessing consumers' decision performance [39], we focus on both subjective and objective assessment of consumers' decision quality and decision effort [1,2]. Specifically, the dependent variables of our investigation are: 1) ***choice confidence***, defined as the extent to which a consumer feels confident in the quality of product choice she has made at an e-commerce website; 2) ***product choice***, which refers to the actual product selection made at the e-commerce website and measured as whether or not the consumer chooses a promoted product (i.e., a product from the sub-set of products promoted by the online store); 3) ***perceived cognitive effort***, defined as a consumer's assessment of the cost associated with evaluating and choosing products at the e-commerce website; and 4) ***extent of product search***, an objective measure of effort, which refers to the total number of products a consumer examines at the e-commerce website prior to making her choice.

2.3. Hypothesis development

Findings from previous behavioral research reveal that consumers are "cognitive misers" who aim to exert as little cognitive effort as possible when retrieving and processing information [40]. To reduce the cognitive processing effort, consumers are known to rely on the wisdom of the crowd when making purchase decisions (e.g., [41–43]). And just as personalized recommendations can be biased to influence consumers' product choices, non-personalized recommendations (generally based on sales volume or consumer ratings) can also be manipulated to mislead consumers. However, we expect that the impact of biased non-personalized recommendations on consumers' decision quality and decision effort will *not* be as pronounced as that of biased personalized recommendations. After all, products liked by others, no matter how good they are, may not be perceived by all consumers as fitting their own product-related needs. The fact that the PRA elicits customers' personal preferences (see Appendix A) adds to the impression that it is making an honest effort to understand and represent consumers' needs. This elevated perception of PRA personalization will increase consumers' identification with and trust in PRAs [44]. Furthermore, in a website providing *biased personalized recommendations*, products are organized such that a consumer can easily observe and ascertain (via intra-list comparison) that a product higher on the recommendation list more closely reflects her preferences (unless the consumer engages in extensive search by examining products appearing in latter pages, in which case better-fitting products may be located). The enhanced perception of personalization not only increases the likelihood that consumers will choose from among the promoted products recommended by the PRA (thus leading to sub-optimal product choices) but also elevates consumers' confidence in their product

Table 2
Measurement items.

	Measures	Adapted from
Choice confidence	■ I am satisfied with the digital cameras that I chose for my friend at the website ■ I am confident that, from the available digital cameras at the website, I selected the best cameras to suit my friends needs ■ These are clearly the best digital cameras available for my friend at the website ■ I would select the same digital cameras for my friend if I had to make the decision again at the website	Pereira [46]
Perceived cognitive effort	■ The task of selecting digital cameras using this shopping advisor was very frustrating ■ The task of selecting digital cameras using this shopping advisor took too much time ■ Selecting digital cameras using this shopping advisor required too much effort ■ The task of selecting digital cameras using this shopping advisor was too complex	Wang & Benbasat [62]
Perceived personalization	■ The website presented products that were personalized to best represent my friend's preferences ■ The website presented products that were tailored to my friend's needs ■ The website presented products that best fitted my friend's requirements.	Komiak & Benbasat [33]

choices.

To exert as little cognitive processing effort as possible, consumers may selectively ignore certain alternatives [45]. In the context of shopping with the assistance of decision aids (e.g., in the form of product recommendations) at e-commerce websites, consumers have been found to rely mostly on the efforts of the decision aids rather than on their own effort in engaging in a thorough decision making process [1,2]. In the context of our study, while both types of biased recommendations are expected to lead to less extensive product search and reduced cognitive effort, consumers receiving biased personalized recommendations may nevertheless be "fooled" into believing that, in addition to making their product searching and selection easier, the PRA does recommend products that best fit their expressed preferences. The heightened level of personalization perceived by consumers may also motivate them to further reduce their own search effort and rely on the PRA's recommendations when making purchase decisions. We thus hypothesize:

H1. a–d: *Type of recommendations exerts significant influence on consumers' decision quality and decision effort. More specifically, compared to those shopping at an e-commerce website providing biased non-personalized recommendations, consumers shopping at an e-commerce website providing biased personalized recommendations will be more likely to choose promoted products (H1a), be more confident in their product choices (H1b), engage in less extensive product search (H1c), and perceive lower level of cognitive effort (H1d).*

H2. a–d: *Perceived personalization mediates the influence of type of recommendations on consumers' product choice (H2a), choice confidence (H2b), extent of product search (H2c), and perceived cognitive effort (H2d).*

3. Research method

An online experiment is conducted to test the effects of different types of recommendations on consumer decision making.

3.1. Independent variables and experimental design

The independent variable, *type of recommendations*, consists of two levels: *Biased personalized* recommendations and *biased non-personalized* recommendations.

Two experimental websites are custom designed for the study. Each website displays the same 96 digital cameras from 8 brands, with 12 products in each brand. The product features for the 12 digital cameras in each brand were carefully designed such that the 6 *promoted* products in a given brand are dominated[3] by the other 6 non-promoted products (i.e., the *dominant* products). That is, each *promoted* product is paired with a *dominant* product in the same brand that has better features but the same price.[4] Hence, each website had 48 *dominant* products and 48 *promoted* products. Each website has 16 product pages, with each page displaying 6 digital cameras. Table 1 and Appendix A illustrate the design for the two experimental websites.

3.2. Measurement instrument

Product choice is operationalized as whether the final selection made by a consumer is a *promoted* product or a *dominant* product, and thus is a binary variable. ***Extent of product search*** is operationalized as the total number of products for which detailed information is viewed by the consumer[5]; it is captured objectively with computer logs. ***Choice confidence***, ***perceived cognitive effort***, and ***perceived personalization*** are measured via 7-point scales adapted from prior research, as shown in Table 2, and validated via several rounds of pilot testing.

3.3. Sample

Participants for this study were 59 e-commerce shoppers recruited from a North American panel maintained by a marketing research firm, which emailed invitations to participate in the study to its members. The firm also provided subjects with a point-based incentive (redeemable for various prizes) for their participation in the study.

3.4. Experimental task and procedures

All participants were randomly assigned to one of the two experimental groups. They were told that a limited selection of digital cameras was currently on sale at an online camera store. They were asked to visit the online camera store, explore the alternatives, and choose a digital camera as a gift for a friend, Pat. They were provided with Pat's camera-related preferences (see Appendix A) and asked to use them to evaluate the PRA's product recommendations[6,7]. Participants were informed that, at the end of the task session, they would be asked a few questions about their shopping experience at the e-commerce website. To motivate participants to take the shopping task seriously, participants were informed before the experiment that the top 25%

[3] Product A dominates product B if A is better in at least one attribute while not being worse in any other attributes compared to product B. In this study, each dominating product offers the same price as but performs better on two important attributes (from among *optical zoom*, *resolution*, *LCD size*, *rapid fire*, and *the number of manual features*) than the pairing dominated product.

[4] We assume that, when product A has better features than product B but is priced the same, consumers are more likely to choose product A.

[5] For each product, only the product's brand, model, and image are shown in a product listing page. Participants can click on the product to reach a detailed product page.

[6] Consumer preferences are often constructed and thus susceptible to the influence of the shopping environment [23]. By providing participants with a friend's profile, we were giving them a relatively stable set of preferences that they could use to evaluate the PRA's recommendations.

[7] Pat's profile was designed such that the products (available at the e-commerce website) best matching Pat's preferences were the *dominant* products, rather than the *promoted* products.

Table 3
Descriptive statistics – individual characteristics.

	Age	Gender	Trust in online shopping	Product expertise	Online shopping experience (in the last 6 months)
Biased non-personalized recommendations	19–29: 6 30–49: 17 50–64: 6	Female: 17 Male: 12	5.241	4.414	Never: 2; Once: 5; 2–4 times: 10; 5–10 times: 5; > 10 times: 7
Biased personalized recommendations	19–29: 6 30–49: 17 50–64: 7	Female: 19 Male: 11	5.354	3.967	Never: 2; Once: 3; 2–4 times: 9; 5–10 times: 7; > 10 times: 9

Trust in online shopping was measured with three items [62]: (1) in general, online stores are trustworthy; (2) in general, online stores keep their promises and commitments; (3) in general, online stores keep customers' best interests in mind. *Product expertise* was measured with two items [62]: (1) I know a lot about digital cameras; (2) among my circle of friends, I'm one of the experts on digital cameras.
The difference between the two groups (reported as a simple average of the individual measures) is non-significant for trust in online shopping ($p = 0.592$) and product expertise ($p = 0.259$).

Table 4
Loadings and cross-loadings of measures.

	Choice confidence	Cognitive effort	Perceived personalization
Confidence1	**0.817**	−0.373	0.670
Confidence2	**0.838**	−0.368	0.494
Confidence3	**0.831**	−0.496	0.603
Confidence4	**0.697**	−0.336	0.569
CogEffort1	−0.485	**0.873**	−0.417
CogEffort2	−0.400	**0.901**	−0.398
CogEffort3	−0.437	**0.917**	−0.325
CogEffort4	−0.336	**0.648**	−0.152
Personalization1	0.611	−0.374	**0.919**
Personalization2	0.650	−0.357	**0.947**
Personalization3	0.783	−0.410	**0.917**

The bold in table highlights the loadings of the measurements of a construct on that construct.

Table 5
Internal consistencies, AVEs, and correlations of constructs.

	Cronbach's alpha	Internal consistency	1	2	3
1. Choice confidence	0.825	0.894	**0.859**		
2. Perceived cognitive effort	0.886	0.929	−0.473	**0.902**	
3. Perceived personalization	0.919	0.949	0.697	−0.427	**0.927**

Note: The scores in the diagonal of the matrix are square root of Average Variance Extracted (AVEs) while the scores in the lower triangle represent the correlations between constructs.

performers in justifying product choices (i.e., how the products chosen satisfy Pat's preferences) would be entered into a draw to get one of twelve $50 gift certificates.

Participants were first asked to complete a short questionnaire to collect demographic data (e.g., age, gender) and background information (e.g., level of experience with computers and online shopping, pre-existing risk perception regarding online shopping). They were then asked to read a tutorial on how to navigate their assigned e-commerce website. Next, they were asked to read a tutorial on digital camera attributes and complete a quiz aimed at testing their understanding of important digital camera attributes. After that, participants were asked to read task instructions and then click on a "Start Shopping" button that would take them to their assigned e-commerce website. Upon the completion of the shopping task, participants were asked to fill out a questionnaire that included the measures of the dependent variables.

Table 6
Results of cross tab analysis: type of product recommendations * product choice.

Type of product recommendations		Product choice		Total
		Dominant product	Promoted product	
Biased non-personalized recommendations	Count	9	20	29
	Expected count	4.9	24.1	29
	% within type of recommendations	**31.0**%	**69.0**%	100.0%
Biased personalized recommendations	Count	1	29	30
	Expected count	5.1	24.9	30
	% within type of recommendations	**3.3**%	**96.7**%	100.0%

Pearson χ^2 (1, N = 59) = 8.038, exact $p = 0.006$.

4. Data analysis and results

Individual characteristics (such as age, gender, pre-existing trust in online shopping, online shopping experience, and product expertise), which could affect consumers' product choices and website evaluations, are controlled via randomization. Analysis of our data also indicates no significant differences among participants in terms of such individual characteristics. Table 3 shows that over 93% of the participants in both experimental conditions shopped at least once during the six months prior to our study, and about half shopped over five times, suggesting that our participants were indeed e-commerce shoppers. Given that 79% of all Americans and 87% of those between 19 and 49 (represented by 79% of our study sample) are now online shoppers and that both men and women are equally likely to shop regularly online (Pew Research Center, 2016), we believe that our sample is a reasonable representation of online shoppers.

4.1. Measurement model

Partial Least Squares (PLS), as implemented in SmartPLS 2.0.M3, is used to assess the measurement properties of the dependent variables

Table 7
Descriptive statistics – dependent variables and mediating variable.

Type of product recommendations	Choice confidence		Extent of product search		Cognitive effort		Perceived personalization	
	Mean	StDev	Mean	StDev	Mean	StDev	Mean	StDev
Biased non-personalized recommendations	5.52	0.90	12.59	16.44	2.96	1.30	5.276	1.195
Biased personalized recommendations	5.95	0.69	6.17	3.88	2.12	0.98	6.089	0.643

Note: The means of *choice confidence*, *cognitive effort*, and *perceived personalization* reported in this table are based on a simple average of the individual measures of each construct.

(except for objective measures) and the mediator (i.e., perceived personalization), all modeled as reflective constructs. Individual item reliability, internal consistency, and discriminant validity are examined following established guidelines [47,48]. All of the loadings of measures (except Confidence4 and CogEffort4, which are excluded in subsequent analysis) with their corresponding constructs exceed 0.7 (Table 4), indicating good item reliability. The composite reliability and Cronbach's alpha of all constructs exceed the threshold of 0.7 (see Table 5), indicating that the measures have good internal consistency.

Our data also satisfy the guideline for establishing convergent and discriminant validity [48]. First, the square root of AVE of each of the two constructs is greater than the correlation of the two constructs (see Table 5). Second, no item loads higher on a construct other than the one it intends to measure (see Table 4).

4.2. Results of hypothesis tests

In this section, we present the results of data analyses conducted to test our hypotheses.

4.2.1. Manipulation check

Analysis of data reveals that consumers' *perceived personalization*[8] is significantly higher in the *biased personalized* recommendation condition than in the *biased non-personalized* recommendation condition (M = 6.089 vs. 5.276,[9] F (1, 57) = 10.684, $p < 0.01$), suggesting that the provision of personalized recommendations (while biased) heightens consumers' perception that the website understands their product related preferences and needs.

4.2.2. Cross tab analysis

A cross-tabulation of *type of recommendations* and *product choice* (see Table 6) reveals that the percentage of participants choosing a promoted product is higher for the *biased personalized recommendations* condition than for the *biased non-personalized recommendations* condition. As one cell has expected count < 5, an exact significance test was selected for Pearson's chi-square. Result shows a significant relationship between the type of recommendations and product choice (χ^2 (1, N = 59) = 8.038, exact $p < 0.01$), with participants receiving *biased personalized recommendations* conditions being significantly more likely to choose promoted products when compared to those receiving *biased non-personalized recommendations*. The effect size of this association is medium (Phi = 0.369) [49]. Therefore, H1a is supported by our data. Further, our finding that 28% more participants in the *biased personalized recommendations* condition chose promoted products compared to the other condition, a finding that is not only statistically significant but also of high practical importance, suggests that our manipulation indeed influenced participants' behavior.

[8] See measurement items for perceived personalization in Table 2.

[9] Although there was no personalization for the biased non-personalized treatment, the website's statement that all the products are ordered by popularity may have contributed to the participants' perception that the products were somewhat personalized to their needs, as popularity is commonly taken as an indicator of quality. It is also likely that some recommended products appealed to the participants as a good fit (i.e., there are no garbage products), thus increasing their perception of personalization.

Table 8
ANOVA results (IV: type of recommendations).

Dependent variable		Sum of squares	df	Mean square	F	Sig.
Choice confidence	Between groups	2.762	1	2.762	4.293	**0.043**
	Within groups	36.666	57	0.643		
	Total	39.428	58			
Extent of product search	Between groups	10.410	1	10.410	7.888	**0.007**
	Within groups	75.225	57	1.320		
	Total	85.636	58			
Perceived cognitive effort	Between groups	607.680	1	607.680	4.329	**0.042**
	Within groups	8001.201	57	140.372		
	Total	8608.881	58			

4.2.3. ANOVA

Table 7 summarizes the descriptive statistics for *choice confidence*, *extent of product search*, and *perceived cognitive effort* for the two experimental conditions.

Results of an ANOVA test (see Table 8) show that *type of recommendations* has significant effect on *choice confidence*, *extent of product search*, and *perceived cognitive effort*. Compared to those shopping at an e-commerce website providing *biased non-personalized recommendations*, participants assigned to an e-commerce website with *biased personalized recommendations* are more confident in their product choices (F(1, 57) = 4.293, $p < 0.05$), engage in less extensive product search (F(1, 57) = 4.329, $p < 0.05$), and perceive lower cognitive effort (F(1, 57) = 7.888, $p < 0.01$). Because effect size has been increasingly used by researchers to evaluate the degree to which findings have practical importance [50,65], we also computed Cohen's d,[10] an effect size statistic for comparing means). The result shows that the effect size for the impact of type of recommendations is medium on choice confidence (Cohen's d = 0.538), medium on extent of product search (Cohen's d = 0.5376), and medium to large on perceived cognitive effort (Cohen's d = 0.7295) [49]. Therefore, our data fully corroborate H1b–d.

4.2.4. Mediation analyses

Mediation analysis was conducted to explore the potential role *perceived personalization* plays in *mediating* the influence of *type of recommendations* on consumers' decision making outcomes. In particular, we aimed to confirm our conjecture that a high level of personalization perceived by participants contributes to their increased likelihood to choose promoted products (indicating suboptimal purchase decision), enhanced choice confidence, reduced product search, and lowered perception of cognitive effort.

The causal steps approach to testing mediation popularized by Baron and Kenny [51], which focuses on examining the individual

[10] Cohen's d is calculated using the effect size calculator for *t*-test available at http://www.socscistatistics.com/effectsize/Default3.aspx.

Table 9
Assessment of indirect effect of type of recommendations (X).

Outcome variables (Y)	Effect	Boot SE	95% LLCI	95% ULCI	R^2_{med}
Product choice	−0.8389	0.7287	−2.2066	−0.1204	
Choice confidence	0.4807	0.1918	0.1974	0.9677	0.0693
Extent of product search	−1.9356	1.3972	−5.0768	0.5152	0.0416
Perceived cognitive effort	−0.2809	0.1472	−0.6001	−0.0153	0.0762

paths in a mediation model, has been heavily criticized on multiple grounds (such as low statistical power and underestimated Type I error) (see [52,53]). In this study, consistent with modern practice in testing indirect influence advocated by Mackinnon, Preacher, Hayes and others (e.g., [52,54,55]), we adopted the more rigorous and powerful bootstrapping method (using Process for SPSS 2.16.3 by Andrew Hayes, with 5000 samples) to assess the mediating role of perceived personalization. Bootstrapping, which is a nonparametric resampling technique, has two advantages over Baron and Kenny's [51] approach [56]: First, indirect effect is measured directly in the bootstrapping method, rather than merely inferred to exist through a sequence of tests. Second, the bootstrapping method imposes no assumption of the normality of the dataset, and thus is recommended for studies with small sample size (as in the case of our study).

The results of our analysis (see Table 9) reveal that *perceived personalization* significantly carries the influence of the independent variable, *type of recommendations* (with biased non-personalized recommendations coded as "0" and biased personalized recommendations coded as "1"), on all the dependent variables (with the exception of *extent of product search*). More specifically, the indirect effects of *type of recommendations* on *product choice* (with promoted products coded as "0" and dominant products coded as "1") (b = −0.8389, 95% CI = −2.2066, −0.1204), *choice confidence* (b = 0.4807, 95% CI = 0.1974, 0.9677), and *perceived cognitive effort* (b = −0.2809, 95% CI = −0.6001, −0.0153) through *perceived personalization* are significant, as none of the three bias-corrected 95% confidence intervals contains zero. Table 9 also shows that an R^2_{med} (R-squared mediation effect size) of 0.0693 for the indirect influence on *choice confidence* and an R^2_{med} of 0.0762 for the indirect influence on *perceived cognitive effort*,[11] suggesting that 6.93% (7.62%) of the variance in *choice confidence* (*perceived cognitive effort*) is attributable to the indirect effect of *type of recommendations* thorough *perceived personalization*. Both point estimates of R^2_{med} are considered small-to-medium as per Cohen's guidelines defining small (0.01), medium (0.09), and large (0.25) effect sizes [57]. Therefore, our data support H2a–b and H2d.

However, contrary to our expectation, the indirect effect of *type of recommendations* on *extent of product search* via *perceived personalization* is not significant (b = −1.9356, 95% CI = −5.0768, 0.5152), as zero is contained in the bias-corrected 95% confidence interval. Our data thus do not support H2c.

5. Discussion and conclusions

This section summarizes the major findings of this study, presents its contributions to research and practice, and discusses its limitations.

5.1. Discussion of findings

Examining how biased web-based product recommendations (particularly biased personalized recommendations from online PRAs) influence consumers' decision making outcomes, this study validates the importance of *perceived personalization* as an important underlying mechanism driving such influence.

Our results show that *type of recommendations* exerts significant influence on consumers' decision making outcomes. More specifically, while consumers receiving *biased personalized recommendations* are more inclined to choose sub-optimal products (an indication of lower objective decision quality) when compared to those receiving *biased non-personalized* recommendations, they will also paradoxically perceive enhanced effectiveness and efficiency of their own decision making, as demonstrated by higher choice confidence, less extensive product search, and lower perceived cognitive effort. In addition, our results reveal that an important mechanism underlying the influence of *type of recommendations* on consumers' actual product choice, confidence in their product choice, and perceived cognitive effort is their perception that the recommendations received are personalized to their individual preferences and needs. An unexpected finding is the non-significant effect of *perceived personalization* in mediating the impact of *type of recommendations* on consumers' *extent of product search*. This suggests that, while biased personalized recommendations influence consumers' product search, they do so directly rather than through the operation of *perceived personalization*.

Additional analysis of our data shows that, 73% of the products viewed by consumers receiving *biased non-personalized* recommendations are promoted products (with 27% being dominant products), suggesting that some consumers in this condition have ventured beyond the first eight pages (on which only promoted products are listed) to check on additional alternatives. In contrast, 93% of products viewed by consumers provided with *biased personalized* recommendations condition are promoted products, indicating a very high level of confidence in the products the PRA recommended. Furthermore, consumers receiving *biased personalized* recommendations have significantly lower perception of the website acting *contrary* to their best interests[12] compared to those receiving *biased non-personalized* recommendations (M = 2.255 vs. 2.879, t = 2.211, p < 0.05). Insomuch as *biased personalized recommendations* lead to impairment in *actual* (i.e., *objective*) *decision quality* (indicated by consumers' choice of sub-optimal products), enhancement in *choice confidence* (i.e., *perceived* decision quality), and lack of awareness of the bias, these findings reveal the serious risk consumers face when shopping with biased PRAs.

5.2. Contributions to research and practice

This study makes significant contributions to research in several ways. Overall, it is the first to our knowledge to illustrate how effecting *perceived personalization* in the context of PRAs could be used as a tactic to increase the acceptance of deceptive recommendations. The findings of this study provide strong evidence for the power of *biased personalized recommendations* in influencing consumers' decision making outcomes. By validating the role *perceived personalization* plays in facilitating such influence, this study has also unraveled the mechanism by which biased personalized recommendations function. More specifically, the results of this study show that *biased personalized* recommendations using PRAs, which result in sub-optimal product choice, nevertheless create the illusion of improved decision quality (an important predictor of consumers' intention to shop at an e-commerce website) as indicated by customers' increased *choice confidence*. This, when coupled with the finding that the influence of biased personalized recommendations on consumer decision making can be explained by the high level of *perceived personalization*, accentuates the danger of the insidious form of manipulation made possible by e-commerce

[11] To the best of our knowledge, the effect size measures for indirect effect are not ready available for mediation models with dichotomous outcomes. Thus, in accordance with the suggestion of Lin, Lucas, and Shmueli [50], we reported the effect size for the indirect effect of type of recommendations on product choice using 95% confidence intervals (see Table 9).

[12] Measured by two items: (1) I fear that ForeverCam.com would act contrary to my best interest, and (2) I am worried that ForeverCam.com is interested in just its own well-being, not in my well-being.

personalization technologies. To our knowledge, this is the first study that draws on the theoretical perspective of personalization to shed light on consumer vulnerability to biased PRA recommendations in e-commerce product websites. Via an empirical investigation of potential danger of personalization systems in e-commerce, this study contributes to a unique perspective on IS research into the dark side of IT (e.g., [3,15,66–69]).

The findings demonstrate that biased personalized recommendations are particularly powerful in influencing consumer decision making at the expense of consumers. It is thus important for government agencies, industry leaders, and consumer organizations to not only promote consumers' awareness of this novel manipulative practice made possible by e-commerce technologies but also educate consumers (via informative brochures or training courses, online and/or offline) on strategies to address and reduce such risks [15]. Additionally, given consumers' vulnerability to biased recommendations from online PRAs, effective interventions (such as, contextualized warning messages) need to be devised to help consumers detect such bias more easily [3].

Furthermore, it is imperative for industry leaders to establish guidelines of good business practices online [15]. For instance, online companies may be required to provide an explanation of the PRA's (or any personalized recommendation systems) reasoning logic [58] and market representativeness [14], provide a fit score for each product recommended and an explanation of how that score is calculated, and/or provide functionalities allowing consumers to sort/search/compare all products by their choice of feature. Strong, enforceable self-regulation and self-policing by industry members are critical to addressing the growing problem of e-commerce manipulation, as unethical merchant behavior affects not only consumers but also honest businesses online. After all, if consumers are reluctant to make purchases online for fear of being swindled, online businesses as a whole suffer from loss in sales and reputation.

5.3. Limitations and future research

This study has some limitations that in turn open up a number of interesting avenues for future research. First, this study focuses on transaction-driven, content-based personalization systems (which offer personalized recommendations based on consumer preferences) [59]. Future research may explore the influence of other types of online personalization systems (see [59]). Second, the study examines only one way by which online PRAs provide biased product recommendations to consumers (i.e., recommending promoted products over products that best fit consumers' preferences), leaving other potential types of manipulation techniques (see [15]) open for investigation. Third, the experimental websites in this study have more products (96 in total) and more product pages (16 pages in total) than those in many other e-commerce studies (e.g., [23,60]). Nevertheless, this is done to simulate consumers' experience at real e-commerce websites, where they typically encounter hundreds of alternatives in a product category. In addition, participants in our study needed to wade through 8 pages of promoted products to reach a dominant product. While we designed Pat's profile such that products best matching Pat's preferences were the dominant products, rather than the promoted products, we acknowledge the possibility of some participants finding (and ultimately choosing) a promoted product that matches their brand preferences perfectly and other camera-related needs closely. Future studies can explore whether (and how) consumers' vulnerability to biased recommendations is influenced by the number of product alternatives and their placement in product pages at e-commerce websites. Fourth, our finding reveals nonsignificant indirect effect of type of recommendations on participants' extent of product search via perceived personalization, which is contrary to what is expected and warrants additional investigation in future research into mechanisms (e.g., conscientiousness) driving participants' product search behavior. Further, while all but one of our hypotheses are supported by data, future research can explore other potential mediating or moderating variables that influence the impact of type of recommendations on consumers' decision making outcomes with a larger sample size (which can increase the power of hypothesis tests).

5.4. Conclusion

Drawing on the theoretical perspective of personalization, this study constructs and tests a research model that examines how consumers' decision making outcomes are influenced by biased product recommendations when shopping online. The results of this study have not only confirmed the effect of biased personalized recommendations on consumers' decision quality and decision effort, but they have also extended prior research by identifying perceived personalization as the underlying mechanism explaining the influence of biased PRAs. With enriched understanding of consumer vulnerability to biased web-based product recommendations (a shadowy but very real phenomenon), this study fills a void in the literature and calls attention to the unsavory side of e-commerce.

Compliance with ethical standards

All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

Conflict of interest

Bo Xiao declares that she has no conflict of interest. Izak Benbasat declares that he has no conflict of interest.

Appendix A. Experimental design

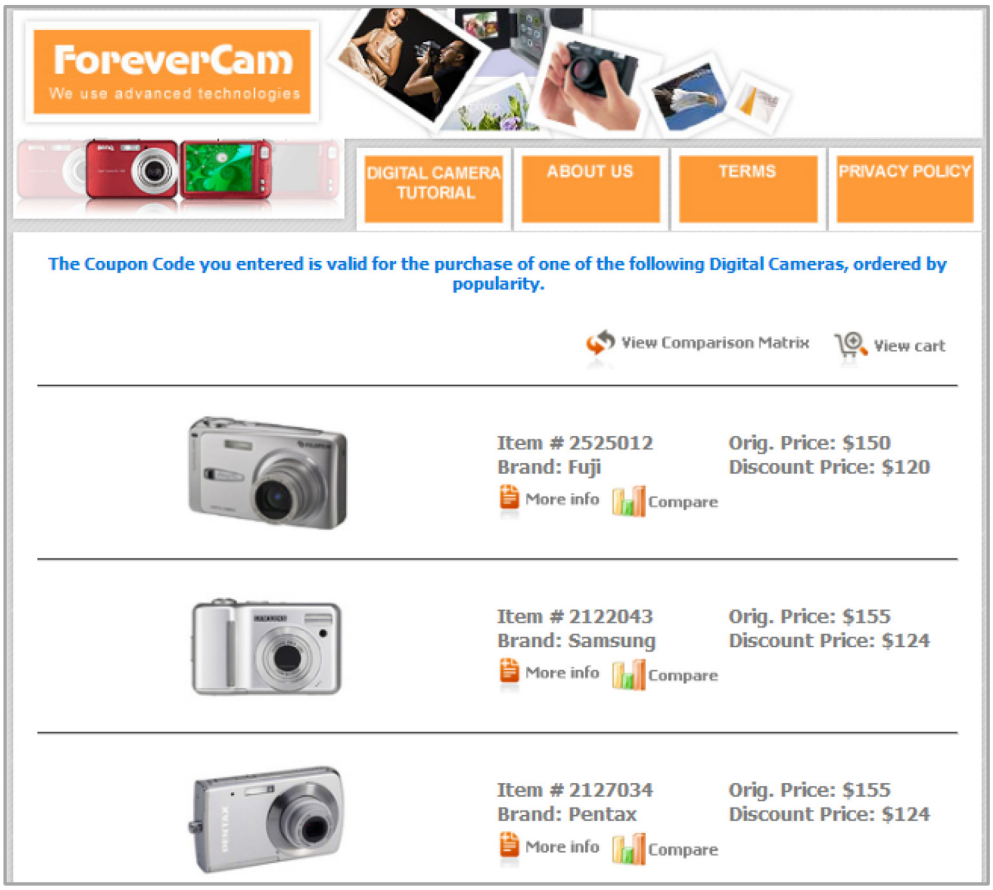
A.1. Pat's camera-related preferences

(Important note: Please use Pat's camera-related preferences (that are presented below), NOT your own preference, to evaluate the shopping advisor's product recommendations.)

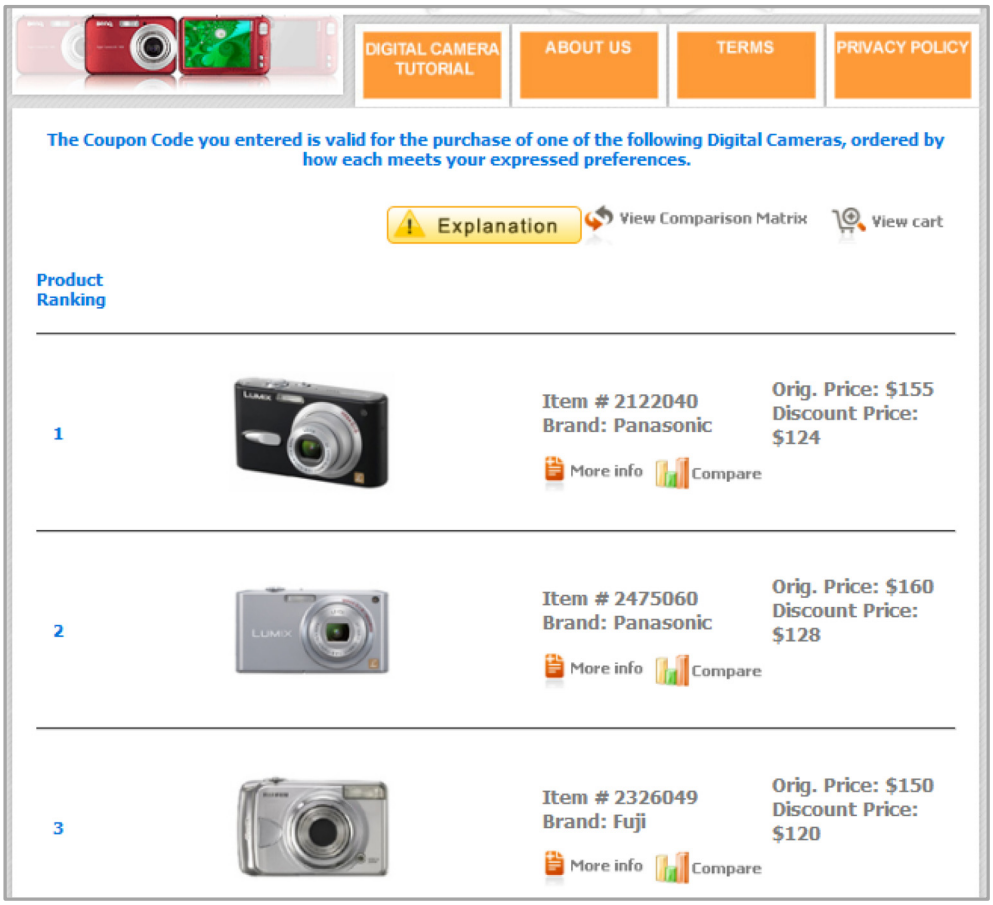
It is extremely important that Pat can use the digital camera to take snapshots of social gatherings, to photograph landscapes and scenery, and to capture action shots at Pat's favorite sporting events (e.g., hockey games and soccer games). Pat wants to find a digital camera that offers enough flexibility with many manual controls. It is also important that the digital camera has a large LCD screen, so that Pat can compose and review pictures easily.

In addition to posting pictures on the web and emailing them to family and friends, Pat also wants to output razor-sharp parings that are 8×10 in. or larger on a high-quality photo printer. Pat only likes digital cameras from Kodak, Samsung, or Fuji.

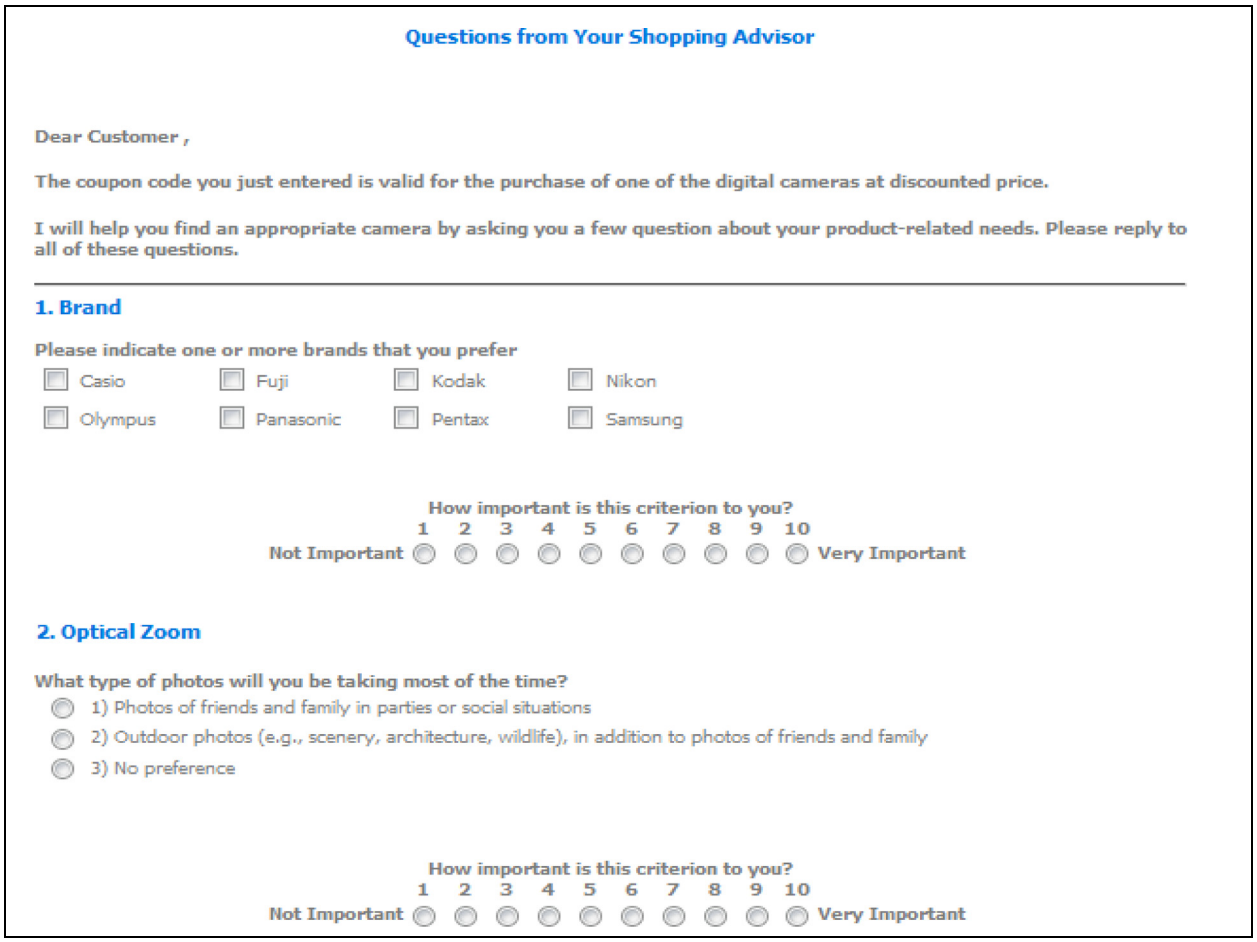
A.2. Website with biased non-personalized recommendations



A.3. Website with biased non-personalized recommendations



A.4. Preference-elicitation interface of the website with biased personalized recommendations



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